Mathematical Logic

Mathematical Logic is the study of mathematics in an abbreviated language, that is, the expression of mathematical concepts in mathematical symbols.

Propositions

A proposition is a declarative sentence (**Statements**) that can be **true** or **false** and cannot be true or false at the same time.

Example

(1) Khartoum is in Sudan (**Ture**)
(2) 1+2 = 5 (**False**)
(3) Help - Help!! (**not proposition**)
(4) Do your homework (**not proposition**)

Notation

We denote the proposition with the letters p, q, r .... And for compound cases $p(p, q, r\ \ldots)$, $q(p, q, r\ \ldots)$.

If p is a true case, write (**T**) and read **True**, and if it is false, write (**F**) and read **False.**

Negation of Proposition

The negation of the proposition p is denoted by $\neg p$ or $\sim p$ and read (**Not *p***). $\neg p$ is true if p is false, and vice versa.

Connectives, Truth Tables

Connectives are used for **making compound propositions**. The main ones are the following (p and q represent given propositions):

Name	**Represented**	**Meaning**
Negation	$\neg p$	not p
Conjunction	$p \wedge q$	p and q
Disjunction	$p \vee q$	p or q (or both)
Exclusive Or	$p \oplus q$	p or q (but not both)
Implication	$p \Rightarrow q$	if b then q
Biconditional	$p \Leftrightarrow q$	p if and only if q

The truth value of a compound proposition depends only on the value of its components. We can summarize **the meaning** of **the connectives** in the following way:

p	q	$\neg p$	$p \vee q$	$p \wedge q$	$p \oplus q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	F	T	T
T	F	F	T	F	T	F	F
F	T	T	T	F	T	T	F
F	F	T	F	F	F	T	T

Tautology, Contradiction, Contingency

1. A proposition is said to be a **tautology** if **its truth value is T** for any assignment of truth values to its components.

Example:

The proposition $p \vee \neg p$ is a tautology.

2. A proposition is said to be a **contradiction** if **its truth value is F** for any assignment of truth values to its components.

Example:

The proposition $p \wedge \neg p$ is a contradiction.

3. A proposition that is **neither a tautology nor a contradiction** is called a **contingency**.

Example

Using **the truth table**, find the truth values in the following

(1) $p \vee \sim p$

p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T
		tautology

(2) $p \wedge \sim p$

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F
		contradiction

(3) $(p \wedge q) \rightarrow (p \leftrightarrow q)$

p	q	$p \wedge q$	$p \leftrightarrow q$	$(p \wedge q) \rightarrow (p \leftrightarrow q)$
T	T	T	T	T
T	F	F	F	T
F	T	F	F	T
F	F	F	T	T
				tautology

(4) $(p \rightarrow q) \vee \sim p$

p	q	$\sim p$	$p \rightarrow q$	$(p \rightarrow q) \vee \sim p$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T
				contingency

Logical Equivalence

When two compound propositions have the **same truth values**, they are called **logically equivalent**.

Note:

The propositions p and q are **logically equivalent** and $p \equiv q$ is written if and only if $p \leftrightarrow q$ is a **tautology**. Otherwise, they are **not logically equivalent**.

Example

(1) $\sim(p \to q) \equiv p \wedge \sim q$

p	q	$\sim q$	$p \to q$	$\sim(p \to q)$	$p \wedge \sim q$	$(\sim(p \to q)) \leftrightarrow (p \wedge \sim q)$
T	T	F	T	F	F	T
T	F	T	F	T	T	T
F	T	F	T	F	F	T
F	F	T	T	F	F	T

(2) $p \to q \equiv \sim q \to \sim p$

p	q	$\sim p$	$\sim q$	$p \to q$	$\sim q \to \sim p$	$(p \to q) \leftrightarrow (\sim q \to \sim p)$
T	T	F	F	T	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

(3) $p \to q \not\equiv q \to p$

p	q	$p \to q$	$q \to p$	$(p \to q) \leftrightarrow (q \to p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

(4) $p \oplus q \equiv \sim p \oplus \sim q$

p	q	$\sim p$	$\sim q$	$\sim p \oplus \sim q$	$p \oplus q$	$(p \oplus q) \leftrightarrow (\sim p \oplus \sim q)$
T	T	F	F	F	F	T
T	F	F	T	T	T	T
F	T	T	F	T	T	T
F	F	T	T	F	F	T

Laws of Algebra of Propositions

1- Idempotent Laws

(i) $p \vee p = p$ (ii) $p \wedge p = p$

2- Commutative Laws

(i) $p \vee q = q \vee p$ (ii) $p \wedge q = q \wedge p$

3- Associative Laws

(i) $(p \vee q) \vee r = p \vee (q \vee r)$

(ii) $(p \wedge q) \wedge r = p \wedge (q \wedge r)$

4- Distributive Laws

(i) $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$

(ii) $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$

5- DeMorgan's laws

(i) $\sim (p \vee q) = \sim p \wedge \sim q$

(ii) $\sim (p \wedge q) = \sim p \vee \sim q$

6- Complement Laws

(i) $p \vee \sim p = \mathrm{T}$ (ii) $p \wedge \sim p = \mathrm{F}$

7- Identity Laws

(i) $p \vee \mathrm{T} = \mathrm{T}$ $p \wedge \mathrm{T} = p$

(ii) $p \vee \mathrm{F} = p$ $p \wedge \mathrm{F} = \mathrm{F}$

8- Double negation

$\sim (\sim p) = p$

9- Implication and its negation

$p \rightarrow q = \sim p \vee q$

Note

All cases for **Laws of Algebra Propositions** can be **proven** by using **truth tables** (definition).

Example

Using the **laws of algebra of Propositions**, prove the following

(1) $\sim (p \vee q) \vee (\sim p \wedge q) = \sim p$

proof

LHS $\sim (p \vee q) \vee (\sim p \wedge q) = (\sim p \wedge \sim q) \vee (\sim p \wedge q)$

$= \sim p \wedge (\sim q \vee q)$

$= \sim p \wedge \mathrm{T} = \sim p$ #

(2) $(p \vee q) \wedge \sim p = \sim p \wedge q$

Proof

LHS $(p \vee q) \wedge \sim p = \sim p \wedge (p \vee q)$

$= (\sim p \wedge p) \vee (\sim p \wedge q)$

$= \mathrm{F} \vee (\sim p \wedge q) = \sim p \wedge q$ #

Open Sentences

The open sentence is **a declarative sentence** with **one** or **several variables**, and it is a simple sentence when each variable in it is replaced by an element of the **substitution set** (**universal set**). The open sentences denoted by $p(x), p(x,y), p(x,y,z)$ ....

The Truth Set

The set of all elements for which the open sentence is true is called the truth set and denoted by p.

Example

Find the truth set in the following

(1) $\mathrm{p}(x):\ x^2-1=0$

$\mathrm{p}=\{1,-1\}$

(2) $\mathrm{p}(x):\ x^2-x-2=0$

$\mathrm{p}=\{-1,2\}$

Quantifiers Symbols

For an open sentence $p(x)$ with variable x,

1- The sentence $\forall x p(x)$ is read "for all $x, p(x)$". The symbol $\forall$ is called the **universal quantifier**.

2- The sentence $\exists x p(x)$ is read "there exists x such that $p(x)$". The symbol $\exists$ is called the **existential quantifier**.

If U is universal set and p is the truth set, then

(1) $\forall \mathrm{xp(x)}$ is **true** if $p=U$ and **false** if $p\neq U$.

(2) $\exists \mathrm{xp(x)}$ is **true** if $p\neq\emptyset$ and **false** if $p=\emptyset$.

Example

Find the value of truth in the following

(1) $(\forall \mathrm{x}\in\mathbb{N}):\mathrm{x}+1\geq 2$

First, we find the truth set p

$\mathrm{p}=\{1,2,3,\dots\dots\}=\mathbb{N}$ $\therefore$ **True**

(2) $(\forall \mathrm{x}\in\mathbb{N}):\mathrm{x}+2\leq 5$

$\mathrm{p}=\{1,2,3\}\neq\mathbb{N}$ $\therefore$ **False**

(3) $(\exists \mathrm{x}\in\mathbb{N}):\mathrm{x}+5<9$

$\mathrm{p}=\{1,2,3\}\neq\emptyset$ $\therefore$ **True**

(4) $(\exists \mathrm{x}\in\mathbb{N}):\mathrm{x}+1=\mathrm{x}$

$\mathrm{p}=\emptyset$ $\therefore$ **False**

Example

If $U = \{1, 2, 3\}$ is an uninvesal set, find the value of truth in the following

(1) $\exists x \forall y : x^2 < y + 1$ ans (True)

(2) $\forall x \exists y : x^2 + y^2 < 12$ ans (True)

(3) $\forall x \forall y : x^2 + y^2 < 12$ ans (False)

(4) $\exists x \exists y : x^2 + y^2 < 12$ ans (True)

(5) $\exists x \exists y \forall z : x^2 + y^2 < 2z^2$ ans (False)

Negation of Quantified Statements

(1) $\sim[\forall x p(x)] = \exists x[\sim p(x)]$
(2) $\sim[\exists x p(x)] = \forall x[\sim p(x)]$

Negation of some symbols

(1) $\sim > \equiv \leq$

(2) $\sim < \equiv \geq$

(3) $\sim \wedge \equiv \vee$

(4) $\sim = \equiv \neq$

Example

Find the negative of the following open sentence

(1) $\forall x : [(x = 1) \wedge (x \leq 3)]$

Solution

$\sim[\forall x p(x)] = \exists x[\sim p(x)]$

$\therefore \sim(\forall x : [(x = 1) \wedge (x \leq 3)]) = \exists x : [(x \neq 1) \vee (x > 3)]$ #

(2) $\exists x : [(x = 0) \to (x > 2)]$

Solution

$\sim[\exists x p(x)] = \forall x[\sim p(x)]$

Let $p \equiv (x = 0)$, $q \equiv (x > 2)$

$$\begin{aligned} \therefore \sim[\exists x : (p \to q)] &= \forall x : [\sim(p \to q)] \\ &= \forall x : [\sim(\sim p \vee q)] = \forall x : [\, p \wedge \sim q] \\ &= \forall x : [\,(x = 0) \wedge (x \leq 2)] \end{aligned}$$ #

Exercise

1- Find the value of truth in the following, then negative the open sentence

(1) $\forall x \in \mathbb{N} : |x| = x$

(2) $\exists x \in \mathbb{R} : x + 1 \leq x$

(3) $\forall x \in \mathbb{N} : x \geq x - 1$

(4) $\exists x \in \mathbb{Z} \ : x + 1 < 1$

(5) $\forall x \in \mathbb{N} : \ x^2 + 2 \geq 4$

(6) $\exists x \in \mathbb{N} : 2x - 5 = 4$

(7) $\exists x \in \mathbb{N} : 10 < x^2 < 15$

2- Find the value of truth in the following if $U = \{1,2,3,4,5\}$, then negative the open sentence

(1) $\exists x : x + 3 = 7$

(2) $\exists x \forall y \ : x + y > x^2$

(3) $\exists x \exists y \ : x^2 + y^2 \geq 30$

(4) $\forall x \exists y \ : \ x + y \leq \ 10$

3- Find the negative of the following open sentence

(1) $\forall x : \{ (x \neq 2) \rightarrow (x \geq 5)\}$

(2) $\exists x : \{ (x = -3) \leftrightarrow (x < 4)\}$

(3) $\forall x : \{[(x = 9) \wedge (x \leq 3)] \rightarrow (x = 0)\}$

4- Check the following propositions are logical equivalences or not:

(1) $p \rightarrow q \ \equiv \ \neg p \vee q$

(2) $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

(3) $\neg(p \longrightarrow q) \equiv p \wedge \neg q$

(4) $p \oplus q \equiv (p \rightarrow \neg q) \wedge (\sim q \rightarrow p)$

(5) $\neg(p \leftrightarrow q) \equiv p \oplus q$

(6) $A \leftrightarrow (B \vee C) \equiv (A \leftrightarrow B) \vee (A \leftrightarrow C)$

(7) $(A \wedge B) \rightarrow C \equiv (A \rightarrow C) \vee (B \rightarrow C)$

5- Using the laws of algebra of propositions, prove the following

(1) $(p \vee q) \rightarrow r = (p \rightarrow r) \wedge (q \rightarrow r)$

(2) $p \rightarrow (q \wedge r) = (p \rightarrow q) \wedge (p \rightarrow r)$

(3) $(p \wedge q \wedge r) \vee (p \wedge q \wedge \neg r) = p \wedge q$

(4) $(p \wedge q) \rightarrow (\neg r \vee \neg s) \ \equiv (r \wedge s) \rightarrow (\neg p \vee \neg q)$